State table:

Flip-Flop: D

//Present(A,B)
0 0
0 0
0 1
0 1
1 0
1 0
1 1
1 1

//Input(x)
0
1
0
1
0
1
0
1

//Next(A,B)
0 0
0 1
0 0
1 0
0 0
1 1
0 0
1 1

//Output(y)
0
0
0

0
0
0
1
1

Kmap_in.txt

3
3
0 0 1 0 0 1 1 0
3
0 1 0 0 0 1 1 0
3
0 0 0 0 0 0 1 1

Kmap_out.txt

#1
2 1 1
1 2 1
#2
2 0 1
1 2 1
#3
1 1 2

Circuit.bench

1 input
1 output
1 inverter
2 flip-flops
6 gates (4 ANDs + 2 ORs)

INPUT(x)

OUTPUT(y)

```
!B = NOT(B)
N1 = AND(x,A)
N2 = AND(x,B)
N3 = AND(x,!B)
DA = OR(N1,N2)
DB = OR(N1,N3)
A = D(DA,CLK)
B = D(DB,CLK)
y = AND(A,B)
#END
```